

Extracting informative features of complex dynamical systems from their pairwise interactions

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Abstract

1 Introduction

A multivariate time series (MTS) can capture the co-evolution of any physical process [9], from the brightness of celestial objects [6], to the movement dynamics of microscopic organisms [14]. Uncovering similar dynamical properties between such vastly different systems can substantially progress scientific understanding. For example, our understanding of the spontaneous synchronization that emerges in diverse physical systems—from flashing fireflies, to Josephson junctions, or social networks [15, 4])—developed by observing and analyzing similar dynamics in coupled phase oscillators, such as the mid-century models of Kuramoto [7] and Winfree [16]. While it is now well-known that certain dynamical properties—e.g., synchronization, long-range correlations, and cascading failures—emerge across many spatiotemporal scales of complex dynamical systems [8], there is still no systematic way to quantify, from data, the similarity between systems that exhibit these properties.

Instead, such similarities are often described qualitatively and referred to as “emergent properties” that arise from the interactions between co-evolving processes [8]. Some paradigmatic examples of dynamical properties emerging from interacting processes include: groups of fireflies suddenly flashing in unison because individuals synchronize with others in their visual field [12]; financial markets crashing due to hidden interactions between asset pairs [13]; and rolling blackouts occurring in electrical grids due to interconnected transmission lines [3]. Although emergence has been established to be a consequence of interactions, it remains elusive as to how we can analyze these interactions in order to quantify similarities in the dynamics of diverse physical processes.

Recent work has demonstrated a way of representing univariate time series in a way that puts diverse time series (of different sampling rates and lengths) in a common space, in which dynamics generated from similar systems are close in the space. This involves representing each time series in terms of a set of extracted real-valued ‘features’, that capture different types of information about dynamical structure, ranging from simple distributional statistics (such as mean and variance) to advanced nonlinear time-series characteristics (such as the correlation dimension and auto-mutual information) [fulcher2013]. Since the diverse feature-vector representation is powerful way to organize, classify, and cluster collections of univariate time series, adapting this approach to distributed dynamical systems that are each measured as a MTS, as depicted in Fig. 1A, could form the basis for powerful new ways to similarly perform statistical learning on MTS. However, devising an appropriate feature-based representation of such data is far more challenging since the types of relevant properties of MTS are vast, from the fundamental level of individual pairs (quantifying how pairs of system elements interact) to the whole systems-level (quantifying the emergent dynamical patterns directly).

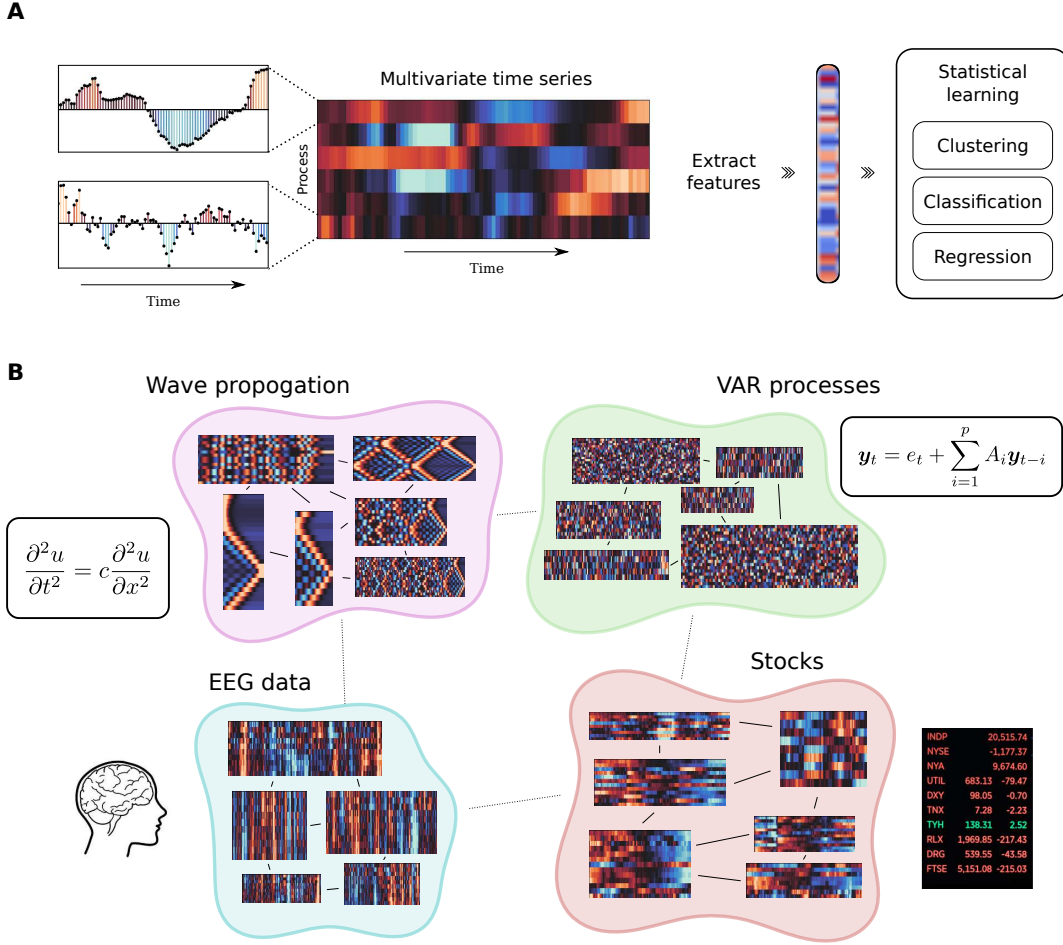


Figure 1: **Complex dynamical systems can be quantitatively compared to one another in order to establish similar behavioral regimes, regardless of the number of processes (time series) or observations (samples).**

One common approach to quantifying the similarity between MTS is to focus on the strength of pairwise dependencies across the system (e.g., ‘functional connectivity’ as a network of Pearson correlations is common in neuroimaging [1]), under the assumption that differences in interactions at the level of specific pairs characterize differences between different types of distributed dynamical processes. However, as different types of processes will in general be characterized by different types of interactions, this approach requires manually selecting an appropriate statistic of pairwise interaction (SPI), that captures the relevant interactions. Furthermore, methods for quantifying and understanding the relationship between pairs of time series are vast, ranging from simple summary statistics (like Pearson correlation) to advanced multi-step causality analysis (like convergent cross-mapping), making the selection far from straightforward. A way forward for tackling this selection problem was outlined in our recent work [2], which unified 250 different SPIs, providing a systematic way to simultaneously leverage diverse scientific methods for quantifying relationships in MTS. This approach is a powerful one for tailoring scientific methods to the needs of a given statistical learning problem involving MTS, allowing researchers to select (or combine sets of) the most appropriate SPIs for a given problem in a data-driven way. However, comparing a collection of MTS in terms of their pairwise dependencies (yielding a $N \times N$ matrix for each $N \times T$ MTS) is only possible for systems with the same number of elements (N), and as it captures how specific system elements relate to other specific elements, requires a common set of elements to be defined across MTS. Furthermore, similarities within (and differences between) general classes of dynamical processes

are characterized by patterns of interactions, rather than the strength of interactions between specific pairs (as captured by the $N \times N$ pairwise dependency representation of MTS). Thus, comparing sets of pairwise dependencies between system elements directly is not an appropriate general representation for organizing MTS according to the similarity of their underlying dynamical processes.

Here we introduce a feature-based representation of MTS that quantitatively captures differences in general equivalence classes of dynamical processes and is applicable to diverse systems of varying sampling rates, sampling periods, and system sizes (cf. Fig. 1B). Our approach leverages different methods for quantifying pairwise interactions between system elements, but instead of representing each MTS directly in terms of the strength of interactions between individual pairs, our approach captures the similarities and differences between the results of applying different SPIs to a given MTS. For example, consider applying three SPIs (Pearson correlation, Spearman correlation, and Transfer Entropy) to a collection of MTS, yielding three pairwise dependencies for each MTS. Our approach would extract features from each MTS as the set of three correlations between the output of three SPIs across the system elements: (i) Pearson and Spearman; (ii) Pearson and Transfer Entropy; and (iii) Spearman and Transfer Entropy. Because the resulting feature vector is always the same size (regardless of the dimensionality of each individual system), this approach puts MTS of different sizes and sampling rates in a common space (allowing diverse MTS to be analyzed using traditional statistical learning algorithms). And because it looks for patterns of similarities between interactions across the full set of system elements, it is able to capture general properties of interactions in a system in a way that is not sensitive to specific pairs of elements. In this work, we introduce this approach for quantifying MTS, and implement it at scale, across a large collection of (> 1000) MTS and combining similarities between a large collection of (> 250) SPIs. We show that this approach provides a robust signature that captures differences between general classes of distributed dynamical processes directly from MTS data. After motivating our approach, we use model simulations of different classes of complex dynamics, we show that our approach successfully organizes MTS according to the underlying class of process, across a wide range of system sizes and scales. Then we use our approach to organize a large database of diverse MTS data (ranging from X to Y to Z [2]), putting a wide range of dynamical systems in a common space for the first time. Our analysis uncovers striking similarities between diverse classes of models and real-world systems from across the theoretical and empirical sciences.

2 Extracting informative features from multivariate time series

The idea underpinning our feature extraction technique is as follows: for certain spatiotemporal dependency structures, a number of statistics for measuring pairwise interactions (SPIs) will yield similar values to one another; for other dependency structures, these same statistics will yield very different values to one another. Therefore, by characterizing the similarity or dissimilarity of a comprehensive set of SPIs, we can in-turn capture a large number of dynamical properties of an MTS.

This basic concept is outlined in Fig. 2, where we illustrate two sets of univariate time-series data from two distinct classes of systems: Gaussian-distributed noise and Cauchy-distributed noise. Moreover, we illustrate how the relationship between the time series is evaluated using two of the most well-known statistics for measuring pairwise relationships (SPIs): Pearson’s product-moment correlation coefficient (Pearson’s ρ), and Spearman’s rank correlation coefficient (Spearman’s ρ_s). As is shown in Fig. 2A, if we were to study the relationships between any two time series generated by sampling a standard Gaussian distribution, using *either* Pearson’s ρ or Spearman’s ρ_s , we would obtain almost identical results, because the assumptions underlying both statistics are valid for Gaussian-distributed data. Instead, as is shown in Fig. 2B, if we were to study the Cauchy-distributed noise with these same SPIs, we would obtain very different results, because the outliers in Cauchy-distributed noise distort Pearson’s ρ but have little effect

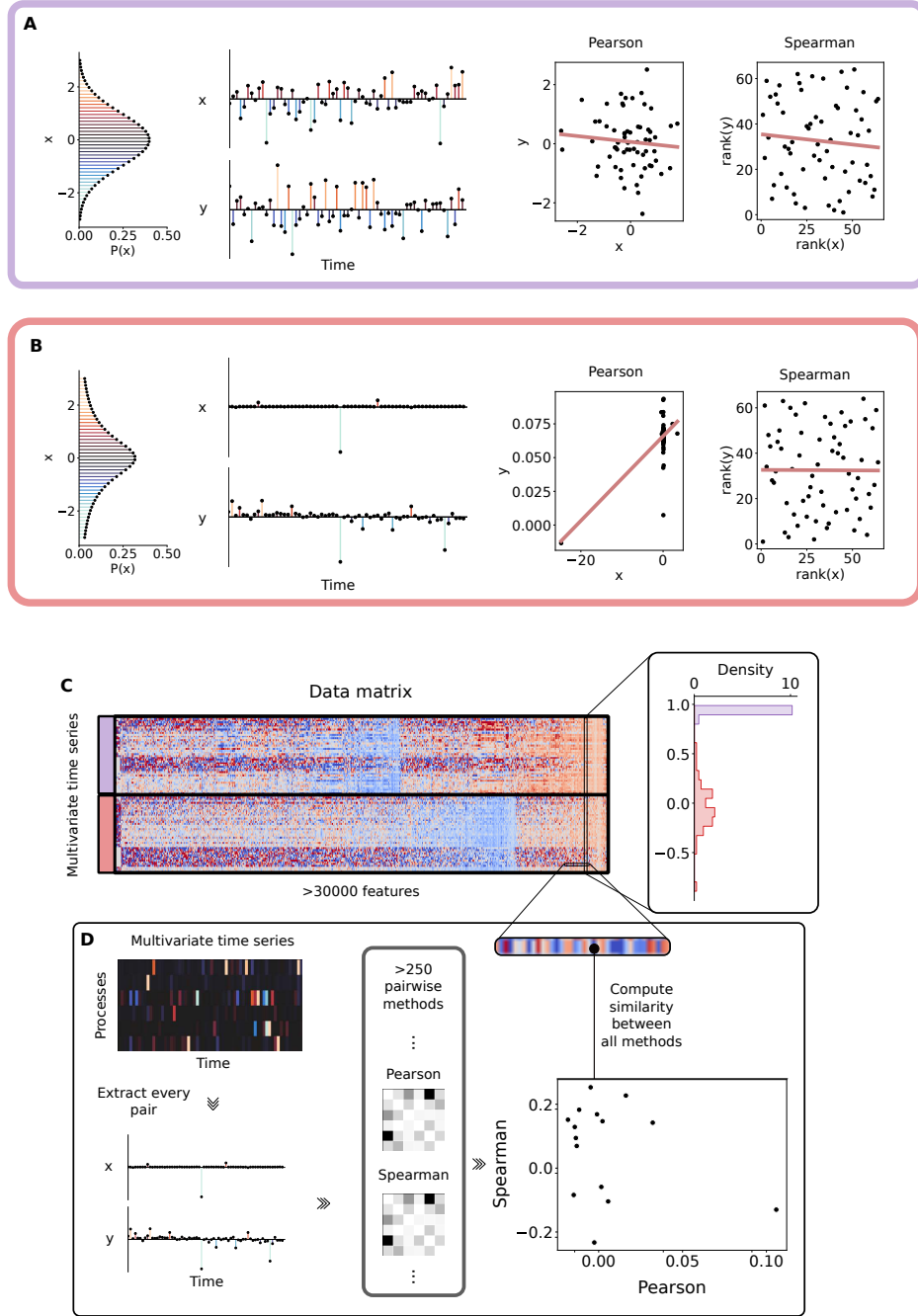


Figure 2: **Different classes of dynamical systems exhibit characteristic patterns of similarities between different statistics for pairwise interactions (SPIs). A B C D**

on Spearman's ρ_s . Thus, one could conclude that the degree to which Pearson's ρ and Spearman's ρ_s agree tells us whether or not the system we are observing is Gaussian- or Cauchy-distributed. While the above intuition works well for simple examples, such as distinguishing two vastly different distributions, the broader notion of distinguishing between two (or more) types of dynamics is not so straightforward.

In general, there are many potential spatiotemporal dependencies that may distinguish one MTS from another, and particular dynamical properties will emerge due to an underlying dependency structure (such as those outlined in Fig. 1). Fortunately, there are also many hundreds of existing techniques (referred to here as SPIs) that have been developed to measure diverse relationships between pairs of time series, including where dependencies are lagged in time (e.g., cross-correlation [10]), may be misaligned (e.g.,

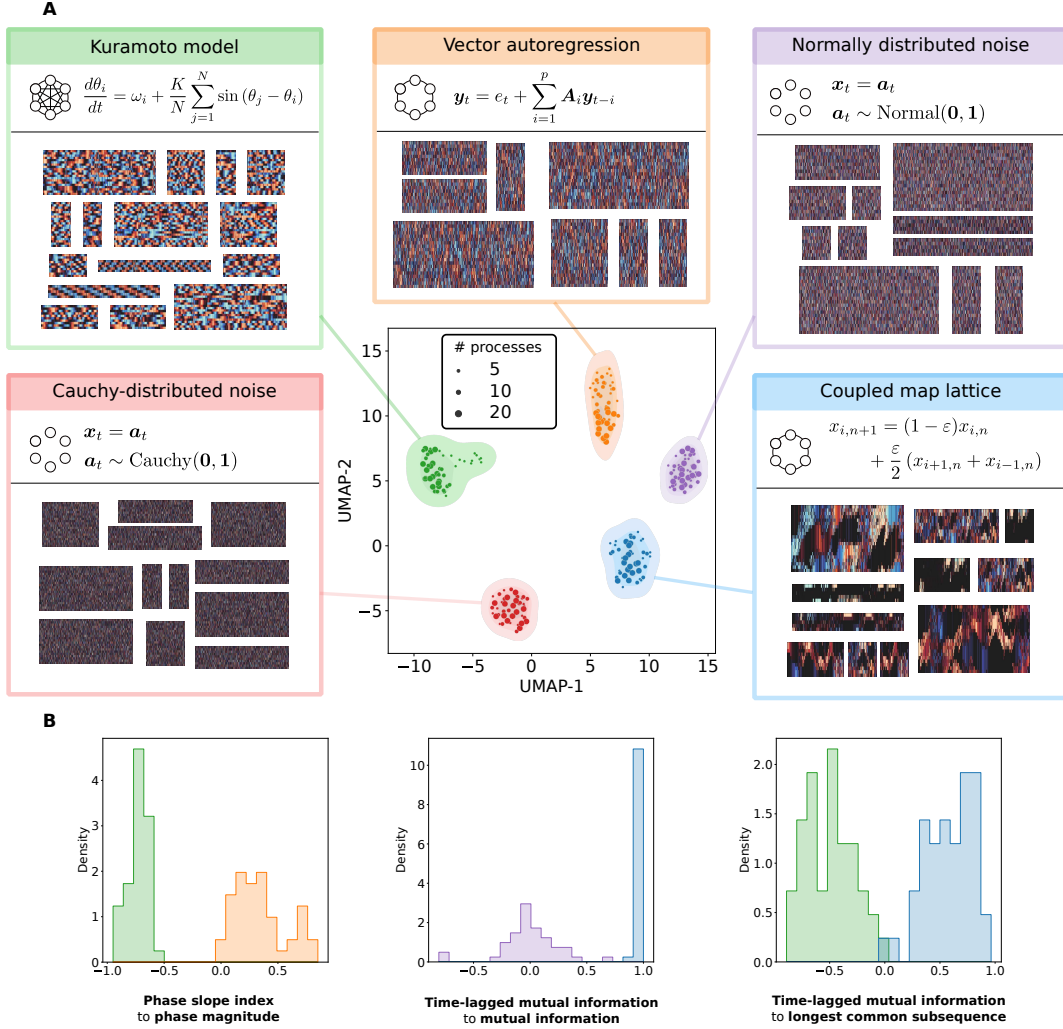


Figure 3: **Extracting informative features from MTS allows us to cluster similar dynamics from data derived from systems of different sizes.**

dynamic time warping [11]), or where the knowledge of one variable improves the predictability of another (e.g., Granger causality [5]). Using our recently developed open-source toolkit, *pyspi* [2], we are able to extract over 250 SPIs from any MTS dataset, which allows us to characterize MTS data based on a diverse range of spatiotemporal dependencies, instead of hand-picking a small number of SPIs (such as the correlation coefficients described above). As illustrated in Fig. 2C–D, our approach to feature extraction from MTS data is to first use *pyspi* to extract hundreds of different types of adjacency matrices from every MTS in a given database. Then, given this comprehensive set of adjacency matrices representing a large number of pairwise interactions occurring within the MTS, we can compare every matrix in order to characterize how strongly they relate to one another. We do so using a “similarity score” derived from Spearman correlation that captures the similarity between every pair of adjacency matrices (for complete details on this score, see [2]). The score between any two adjacency matrices is then one informative feature of the MTS data, in the same way that the similarity between Pearson and Spearman correlation differentiates Gaussian- from Cauchy-distributed noise. Given we have > 250 adjacency matrices, we obtain $> 250(249)/2 = 31125$ features in our feature vector to characterize an individual MTS. Crucially, regardless of the dimensionality of the MTS dataset (i.e., number of processes or observations), we will obtain the same size feature vector because its length is determined by the number of SPIs rather than properties of the data. Finally, we combine the matrix of feature vectors for every MTS in our universe into

a data matrix, shown in Fig. 3C, which can then be used in statistical learning tasks, such as clustering, classification, or regression.

3 Clustering high-dimensional mathematical models using features

As a proof of concept of our feature extraction method, we extend our database to include five different classes of data: Cauchy-distributed noise, Gaussian-distributed noise, vector autoregressive (VAR) processes, Kuramoto oscillators, and coupled map lattices (CMLs). Moreover, following from our problem definition, our aim is to ensure our method can extract robust features of datasets of different scales. As such, we generated 18 datasets per class, where each dataset within a given class is generated using the same parameters, but with both the dimensionality of the MTS data (comprising either 5, 10, or 20 time series per dataset) and number of samples (having either 250, 500, 750, 1000, 1250, or 1500 observations) varied. For each of these datasets, we extract one feature vector per dataset using the method outlined above, forming a data matrix from the 90 feature vectors (18 datasets for 5 classes) from which we can perform statistical learning.

The datasets and results of our feature extraction are illustrated in Fig. 3, where our method is able to cluster the same classes of data together, regardless of their number of observations or processes. To illustrate the separation between the feature vectors, we use UMAP to reduce the high-dimensional data matrix to two dimensions that illustrate clear clusters of MTS data separated by class, irrespective of the number of processes or observations for each dataset. In this low-dimensional space, there is a strong separation between each of the classes, indicating distinct clusters in the high-dimensional (full feature) space. The separation of distinct classes illustrates that pairwise interactions can be used to extract informative features of a consistent size from any MTS data.

Moreover,

4 Unveiling hidden relationships between multivariate time series

Having now established that our method works well for a small set of generative models, we will now apply the same technique to establish similarities amongst a larger database of many different types of dynamics.

5 Discussion

6 Acknowledgments

William Gilpin you legend.

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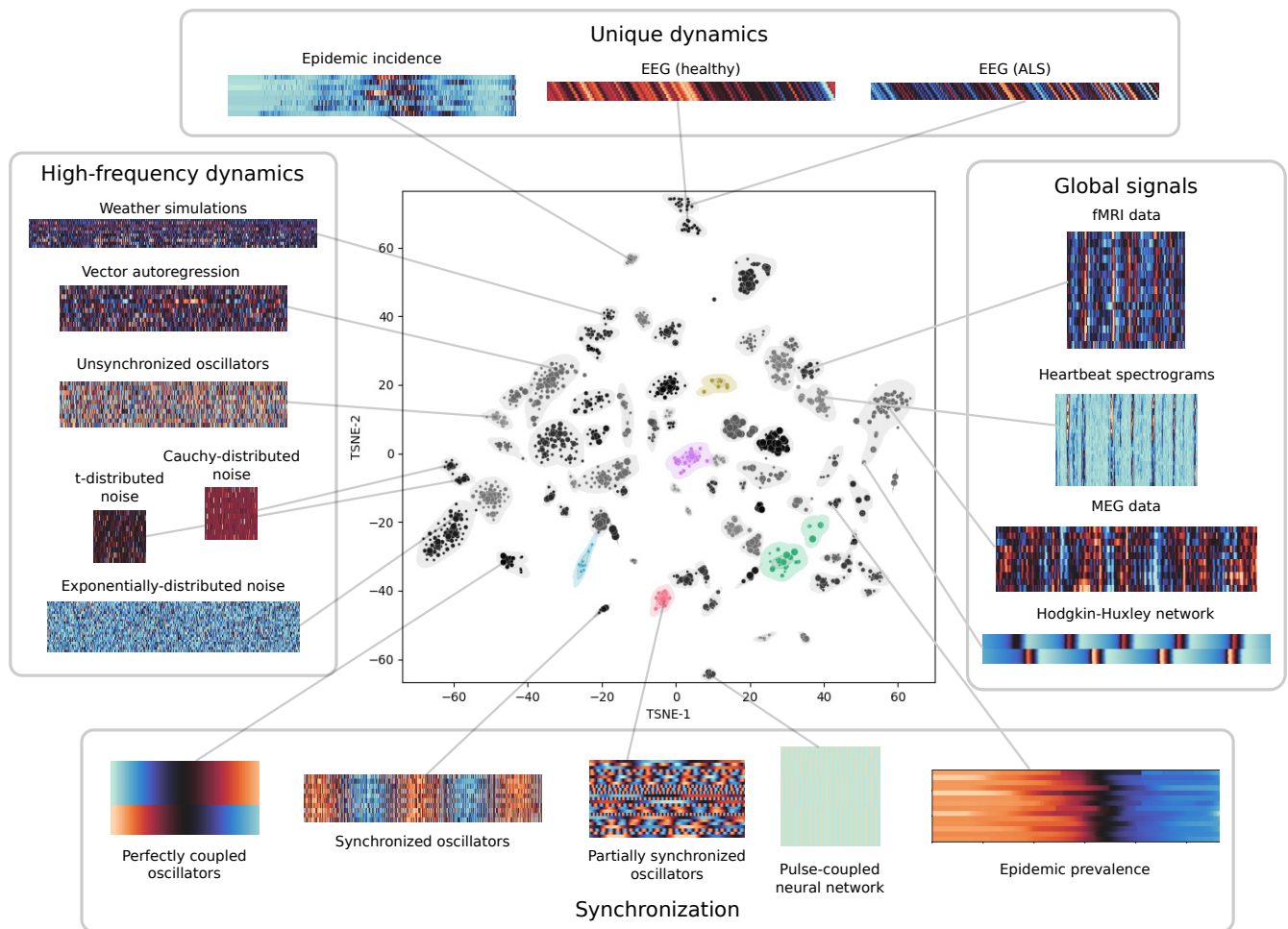


Figure 4: **Feature extraction can be used to structure diverse time series into a coherent space.**

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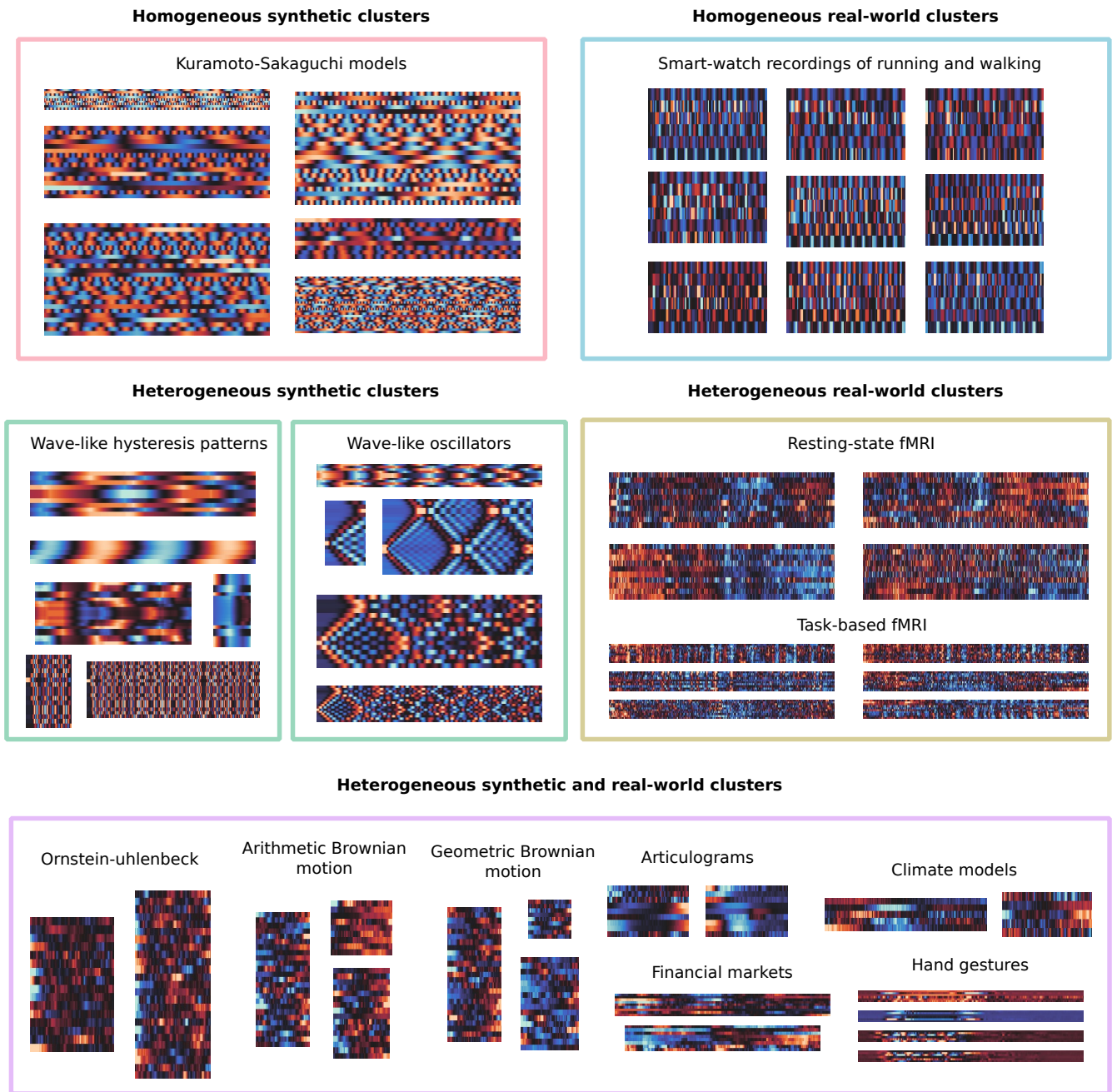


Figure 5: Clusters represent qualitatively similar systems, whether they are from the same model or data type (homogeneous) or from completely distinct systems (heterogeneous).

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